

Project Title

E-commerce Platforms for Traditional Handicrafts from Southern Vietnam

I. Criterias in phase 2

- User-Centered Design: Design intuitive and user-friendly interfaces for both artisans (for product listing, order management, etc.) and customers (for browsing, purchasing, etc.), considering varying levels of digital literacy. 3.2.2
- E-commerce Platform Development: Develop robust and secure e-commerce platforms based on the defined technical requirements, utilizing appropriate technologies and adhering to best practices in web development and security.
- Secure Payment Gateway Integration: Integrate secure and reliable online payment gateways that are commonly used in Vietnam and are accessible to both local and potentially international customers.
- Logistics and Shipping Integration: Develop or integrate solutions for managing product shipping and logistics, potentially partnering with local shipping providers and offering various shipping options.
- Multi-Language Support Implementation: Implement multi-language support, prioritizing Vietnamese and potentially English or other relevant languages.
- Mobile Responsiveness: Ensure that the e-commerce platforms are fully responsive and accessible on various mobile devices.

II. Research information

1. What **technology platforms** is Kien Giang Province using to improve the **artisan communities**? (Phuc)

Kiên Giang Province has been developing its artisan communities by promoting handicraft production, particularly using water hyacinths to create baskets, hats, carpets, and other products. The province has established co-operative groups, co-operatives, and weaving clubs to boost sales and provide stable incomes for artisans

Additionally, Kiên Giang is focusing on attracting investment in high-tech industries, including science, technology, and innovation, which could support artisan communities through modern tools and digital platforms

2. How does **the digital literacy** level of artisan communities in Kien Giang Province? (Hieu)

- . According to the Department of Education and Training, numerous schools struggle with unstable internet connectivity and lack adequate online learning equipment.

- In particular, from the new school year 2024 - 2025, the Department of Education and Training has applied information technologies in teaching and learning through the Khan Academy Vietnam (KAV) Open School model. By using the Khan Academy platform, teachers can assign assignments and evaluate students quickly. Students, especially those in island areas, can study online anytime and anywhere without fear of interruption.

3. What **latest technologies** has Kien Giang Province applied in the year 2024? How were they impact? (Hoang)

a. Digital Transformation in Education: KAV Open School

The Department of Education and Training in Kien Giang decided to use Khan Academy Vietnam (KAV) Open School to support digital education. As a result of this initiative, teachers could quickly handle both student work and assessment and students from island areas could continue learning online without interruption. Launched with great anticipation, the programme has played a key part in advancing digital learning in the province and improving school access.. [The World and Vietnam Report](#)

b. E-Government Initiatives: E-Office Platform

Together with VNPT, the province developed the Electronic Office (E-Office) platform which helped update how offices operate. Thanks to this system, locations operated in a paperless manner, documents were managed efficiently and communications improved within government agencies, all contributing to the development of broader e-government. [VCCI Business Forum+2Vinaphone+2vnpt.com.vn+2vnpt.com.vn+1Vinaphone+1](#)

c. Advancements in Agricultural Practices

- **Organic Rice Cultivation Expansion:** The province is growing organic rice and by 2025 they want to use 6,500 hectares across 11 communes before expanding to 25,000 hectares by 2030. Its aim was to make animal farming more sustainable, add worth to farm products and help protect the environment. [expo.vn+2Vietnam+\(VietnamPlus\)+2english.vtv.vn+2english.vtv.vn](#)
- **High-Quality, Low-Emission Rice Project:** They began work on a scheme that will see one million hectares managed sustainably and produce low emission rice by 2030. Experts applied advanced techniques to decrease rice pollution, use resources more efficiently and follow green

development goals. [expo.vn+1Vietnam Agriculture+1](#)

3. What are the **limitations** of their platforms? (Hoang)

- **Infrastructure Constraints:** The use of KAV Open School showed challenges because of unreliable internet and a shortage of computers in island schools. [The World and Vietnam Report](#)
- **Digital Infrastructure Development:** Although 5G networks are being piloted in chosen sections of the country, Bangladesh still needs to fix issues with complete digital coverage and stable internet connectivity. [VCCI Business Forum](#)
- **User Adoption and Training:** To get the best use out of e-government platforms, employees in government need regular training and updates. [VCCI Business Forum+2Vinaphone+2vnpt.com.vn+2](#)
- **Agricultural Transition Challenges:** Implementing organic and low-emission rice farming means farmers must change their ways, so they need guidance, available resources and help to succeed. [english.vtv.vn+2expo.vn+2Vietnam+ \(VietnamPlus\)+2](#)

III. Interface

Okay, this clarifies the scope significantly: you're thinking about a multi-platform ecosystem with specific functionalities tailored to each interface.

Here's a comprehensive breakdown of the basic functions required for each component, incorporating all previous discussions about the Kien Giang artisan communities, the unified "User/Buyer" interface, and your group's administration role:

1. Users/Buyers App (Mobile Application)

This app focuses on seamless Browse, purchasing, and managing buyer-centric activities.

1. Authentication & Account Management:
 - User Registration/Login: Simple and secure registration using email/password or social logins (e.g., Google, Facebook).
 - Profile Management: View/edit personal details, shipping addresses, saved payment methods.
 - Order History: View past and current orders, including status updates.
 - Wishlist/Favorites: Save products for later.
 - Password Management: Forgot/reset password functionality.
 2. Product Discovery & Browse:
 - Home Feed: Personalized or curated display of products, featured artisans, Kien Giang craft highlights.
 - Category Browse: Intuitive navigation through Kien Giang-specific craft categories (e.g., Sedge Weaving, Khmer Textiles, Woodwork, Local Delicacies).
 - Search Functionality: Efficient search bar with suggestions.
 - Filtering & Sorting: Refine results by price, artisan, location (district in Kien Giang), materials, newest, popular.
 - Detailed Product Pages: High-resolution images (zoom/swipe), detailed descriptions, materials, dimensions, artisan story, reviews, and "Add to Cart" button.
 - Artisan Profiles (View Only): Ability to view an artisan's profile page, their story, other products, and location.
 - Customer Reviews & Ratings: View existing reviews, ability to submit reviews post-purchase.
 3. Shopping & Checkout:
 - Shopping Cart: Add/remove items, update quantities, view subtotal.
 - Seamless Checkout Flow: Guided steps for shipping information, payment selection, and order review.
 - Multiple Payment Options: Integration with popular local payment gateways (e.g., bank transfer, e-wallets like Momo, ZaloPay, credit/debit cards).
 4. Communication:
 - Notifications (Push/In-App): Order updates, new product alerts, promotional messages.
 - Contact Support: Easy access to contact your administration group.
 - Direct Messaging (Optional): Secure chat with artisans for pre-purchase inquiries or custom orders.
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2. Artisan Communities App (Mobile Application)

This app empowers artisans to manage their shop and orders directly from their mobile devices.

1. Authentication & Dashboard:
 - Artisan Login: Secure access to their dedicated artisan dashboard.
 - Dashboard Overview: Quick summary of new orders, pending items, recent sales, and notifications.
2. Profile Management:
 - Artisan Profile Editor: Upload/edit bio, profile picture, contact details, Kien Giang location, and craft specialization.
 - Portfolio/Gallery: Upload photos/videos of their workshop, craft process, or non-listed works.
3. Product/Listing Management:
 - Add New Product: User-friendly form to upload product details (name, description, price, categories, Kien Giang specific tags, images/videos).
 - Edit/Delete Existing Products: Update product information, pricing, stock levels.
 - Product Status Control: Mark products as "active," "draft," "sold out," "on hold."
4. Order Management:
 - View New Orders: Notification and list of incoming orders.
 - Order Details View: Access buyer information, order items, shipping details.
 - Update Order Status: Mark orders as "processing," "shipped," "completed," "cancelled."
 - Shipping Label Generation/Info: (Optional, depending on shipping integration).
 - Accept/Decline Custom Order Requests: If applicable.
5. Communication:
 - Inbox/Messaging: Secure communication with buyers regarding specific orders or inquiries.
 - Notifications: Alerts for new orders, messages, or platform updates.

3. Administration (Web-Based Platform - for your group)

This is the central control panel for your group to manage the entire ecosystem. While it *could* have an app component, the web platform is essential for comprehensive management.

1. Secure Administrator Authentication:

- Admin Login: Secure login with role-based access.
- Multi-Level Permissions: Assign different roles within your group (e.g., Super Admin, Moderator, Content Manager) with specific access rights.
- 2. Centralized Dashboard:
 - Overview of Key Metrics: Total sales, active users, pending artisan approvals, new orders, site performance.
- 3. User Management (Unified):
 - View & Manage All Users: Comprehensive list of all "Users" (buyers) and "Artisans."
 - Artisan Onboarding & Approval Workflow: Review new artisan applications, approve/reject, set up their shop.
 - Account Actions: Edit user profiles, suspend/activate accounts, reset passwords.
- 4. Content & Listing Moderation:
 - Product Listing Approval Queue: Review and approve all new/updated product listings submitted by artisans to ensure quality and adherence to guidelines.
 - User/Artisan Content Moderation: Manage reviews, comments, and artisan profile content.
 - Category & Tag Management: Create, edit, and organize product categories and searchable tags relevant to Kien Giang crafts.
- 5. Order & Transaction Oversight:
 - Full Order List: View all orders placed on the platform.
 - Order Details: Access complete details of any order (buyer, artisan, products, payment, shipping).
 - Refunds & Cancellations: Process and manage refund requests and order cancellations.
 - Dispute Resolution Tools: Mediate conflicts between buyers and artisans.
- 6. Website Content Management System (CMS):
 - Static Page Editor: Easily create and update pages like "About Us," "FAQ," "Privacy Policy," "Terms of Service."
 - Blog/News Editor: Publish and manage articles about Kien Giang's craft heritage, artisan stories, and platform news.
 - Banner/Promotional Management: Control featured sections, banners, and promotional campaigns on the user app.
 - "About Kien Giang" Content Management: Dedicated section to update information about the province and its unique craft communities.
- 7. Reporting & Analytics:

- Sales Reports: Detailed reports on overall sales, sales by artisan, top-selling products, revenue trends.
 - User & Artisan Activity Reports: Insights into user engagement, artisan performance, popular search terms.
 - Traffic Analytics: Integrate with tools like Google Analytics to track website and app usage.
8. System Settings & Configuration:
- General Site Settings: Manage platform name, logo, contact info, social media links.
 - Payment Gateway Configuration: Set up and manage payment methods.
 - Shipping Rule Management: Define shipping zones, pricing, and carrier options.
 - Email Template Customization: Edit automated notification emails for all user types.

IV. Database: Using SQL

V. Summary ideas for phase 2:

Description: There will be three main interfaces: one is for the Administration - our team, User/Buyer, and Artisan Communities. Additionally, three of them will be programmed for both website and mobile phone app. However, for easy way, I decide to implement the website first, app will be in the future.

- For storing database: SQL Language

- For User/Buyer interface: Basic function like e-commerce web in general. However, I want add something unique functions like filtering and sorting based on the price, most selling, comments, star,.....

- For Artisan communities interface: The interface will include latest news in general in vietnam and artisan communities based on the location, trading (between communities to enhance collaboration and relationship), and partners (local and global).

=> Creativity implement: Default language is Vietnamese, but if foreigners want to access, change to English, Auto matching need & supply

=> Code if I can and higher grade

Some core platforms that AI recommend:

+ Core Platform Functionality

+ Unique Features for Artisan E-Commerce in Kien Giang Province.

Need to choose one of concept.

V. Tasks:

- What technologies do team need in coding? (Just research and list their benefits). If there is another solutions, go for it. (Hiếu)

Ans: Technologie use in coding website for front-end and back-end.

- + Front-end: HTML, CSS, Javascript, and important libraries.
- + Back-end (Storing data): MySQL
- What technologies do e-commerce platforms use to secure their payment?

(Phúc)

+SSL/TLS Encryption

+Payment Gateway

+Tokenization

+Two factor Authentication

Ans: E-commerce platforms use a variety of security technologies to protect customer data and ensure secure transactions. These include encryption, firewalls, antivirus software, biometric authentication, and digital signatures. Additionally, AI and machine learning are being used to detect patterns and anomalies, and identify fraudulent activities.

- What technologies do they use to prevent unwanted risk? (Hoàng)

Ans: E-commerce platforms employ various technologies to mitigate risks, including firewalls, encryption, antivirus/anti-malware software, and AI-powered fraud detection systems. SSL certificates ensure secure data transmission, and compliance with PCI DSS standards safeguards customer data. Additionally, two-factor authentication and robust access management policies enhance security.

- What do artisan communities need right now?

Ans: Artisan communities currently need support to bridge gaps in accessing capital, market recognition, and sustainability. They also require assistance in preserving cultural heritage and promoting fair wages.

- What do artisan communities expectation?

Artisan communities often have expectations related to preserving their cultural heritage, promoting sustainable practices, and ensuring fair economic benefits for their members. They also expect support for skill development and access to markets.

*Recommend: Should research at least two e-commerce platforms (like Shopee, Amazon,....)

-> Deadline for doing tasks: 1st June (Monday) 23:59PM

